

Introduction to Direct Access Files.

Sequential Files - Adding a Record

Ordered

Rec001
Rec005
Rec045
Rec090
Rec100
Rec150
Rec300
Rec325

^Z

Rec085

Unordered

Rec300
Rec150
Rec005
Rec090
Rec045
Rec100
Rec001
Rec325

^Z

Sequential Files - Adding a Record

New-Ordered

Rec001
Rec005
Rec045
Rec085
Rec090
Rec100
Rec150
Rec300
Rec325
^Z

Rec085

Extend-Unordered

Rec300
Rec150
Rec005
Rec090
Rec045
Rec100
Rec001
Rec325
Rec085
^Z

Sequential Files - Deleting a Record

Ordered

Rec001

Rec005

Rec045

Rec090

Rec100

Rec150

Rec300

Rec325

^Z

Rec150

Unordered

Rec300

Rec150

Rec005

Rec090

Rec045

Rec100

Rec001

Rec325

^Z

Sequential Files - Deleting a Record

New-Ordered

Rec001
Rec005
Rec045
Rec090
Rec100
Rec300
Rec325
^Z

Rec150

New-Unordered

Rec300
Rec005
Rec090
Rec045
Rec100
Rec001
Rec325
^Z

Sequential Files - Amending a Record

Ordered

Rec001

Rec005

Rec045

Rec090

Rec100

Rec150

Rec300

Rec325

^Z

Rec045

Unordered

Rec300

Rec150

Rec005

Rec090

Rec045

Rec100

Rec001

Rec325

^Z

Sequential Files - Amending a Record

New-Ordered

Rec001
Rec005
Rec045
Rec090
Rec100
Rec150
Rec300
Rec325
^Z

Rec045

New-Unordered

Rec300
Rec150
Rec005
Rec090
Rec045
Rec100
Rec001
Rec325
^Z

Relative Files - Organization

Relative
Record
Number



Relative Files - Adding a Record

Relative
Record
Number



Rec327

Relative Files - Adding a Record

Relative
Record
Number



Rec327

Relative Files - Deleting a Record

Relative
Record
Number



Rec325

Relative Files - Deleting a Record

Relative
Record
Number



Rec325

Relative Files - Amending a Record

Relative
Record
Number



Rec007

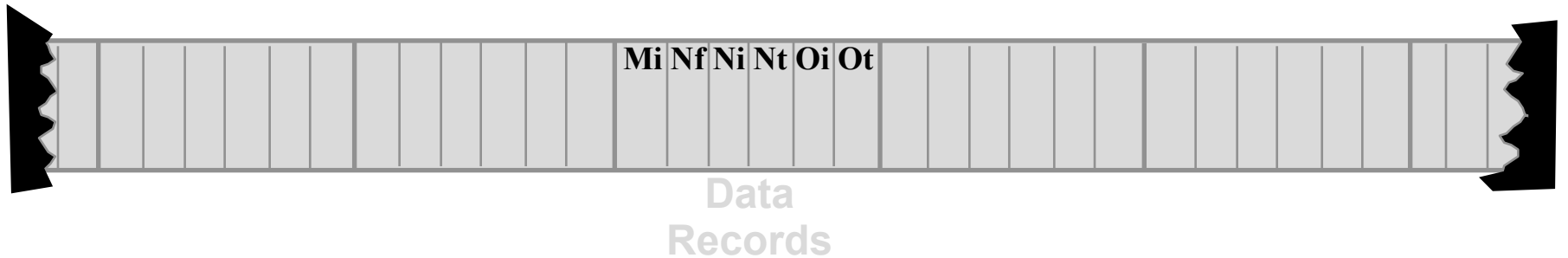
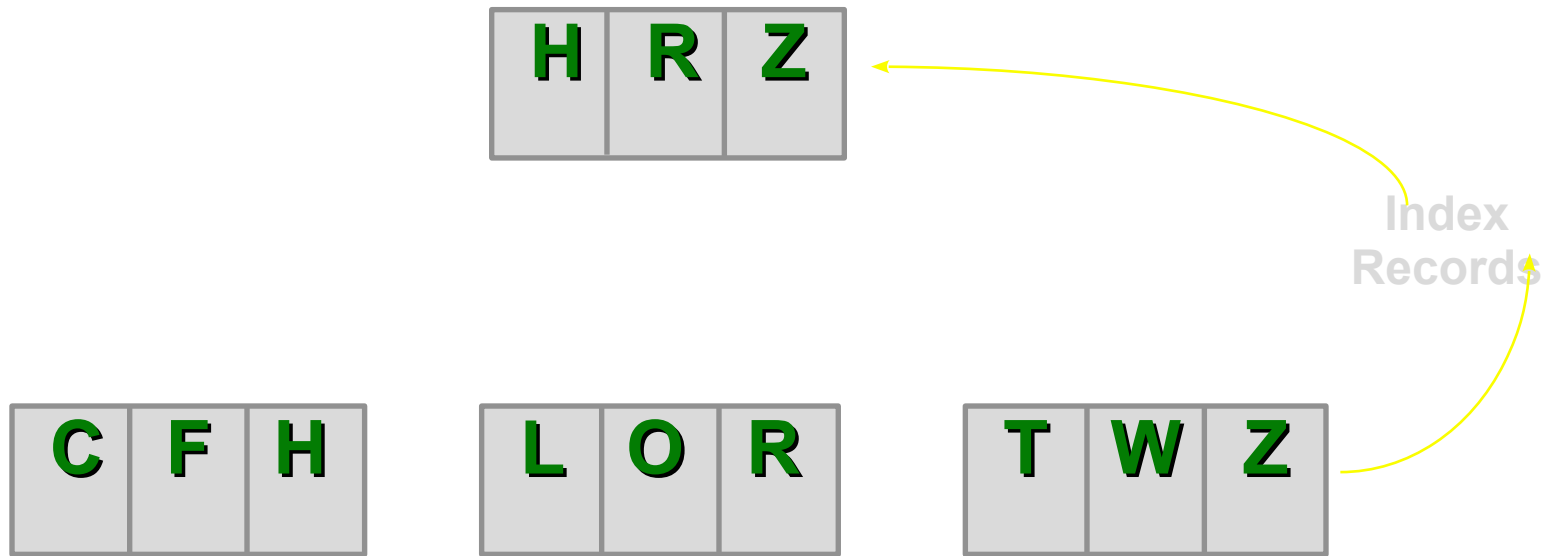
Relative Files - Amending a Record

Relative
Record
Number

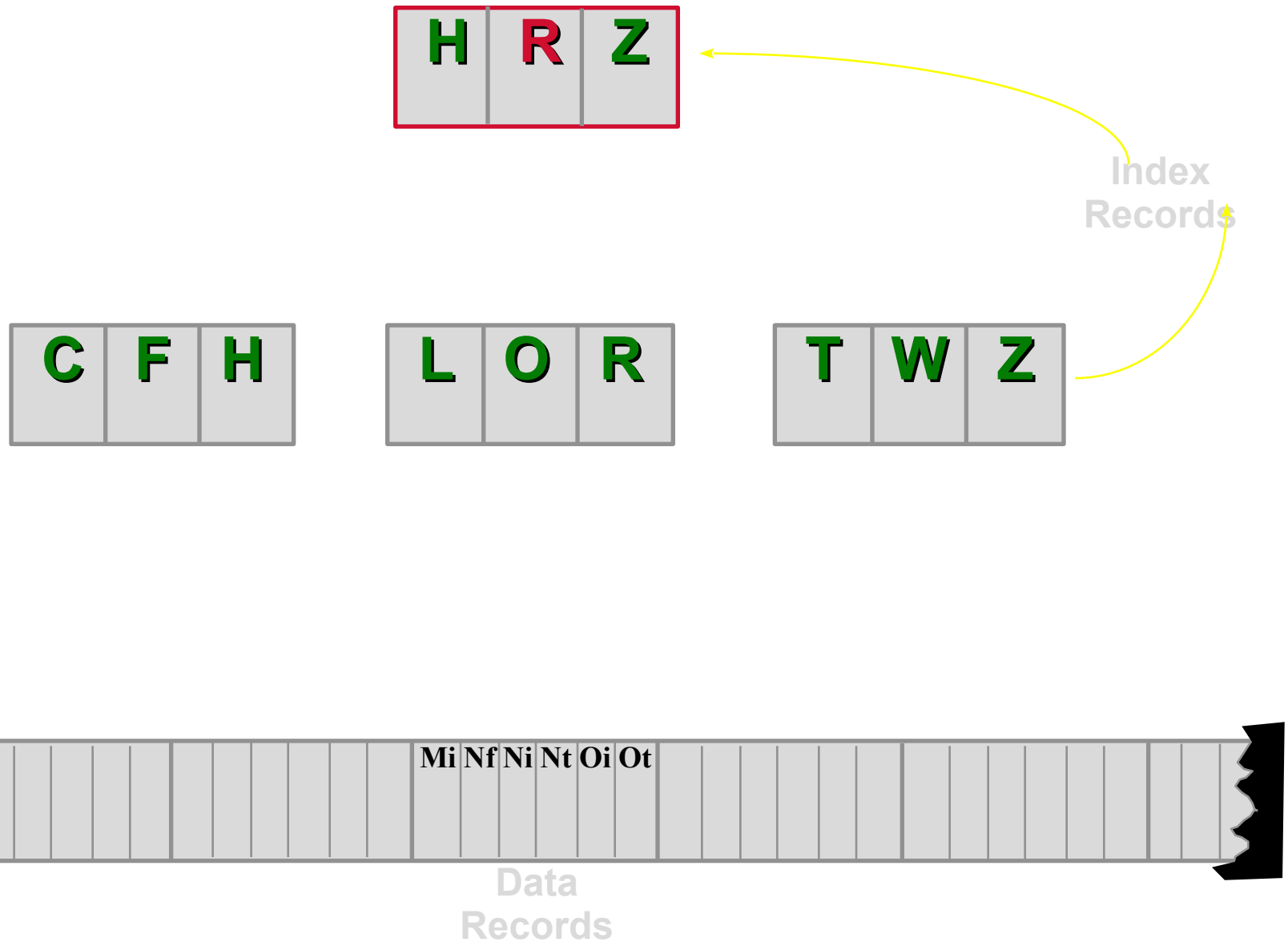


Rec007

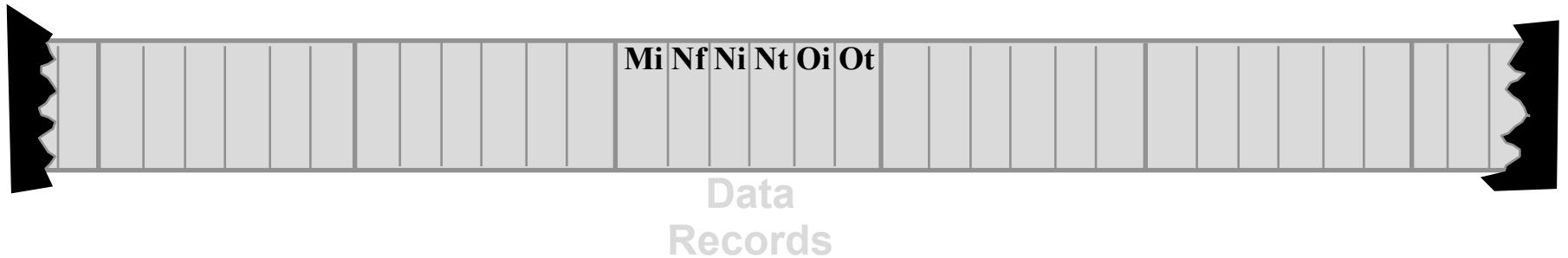
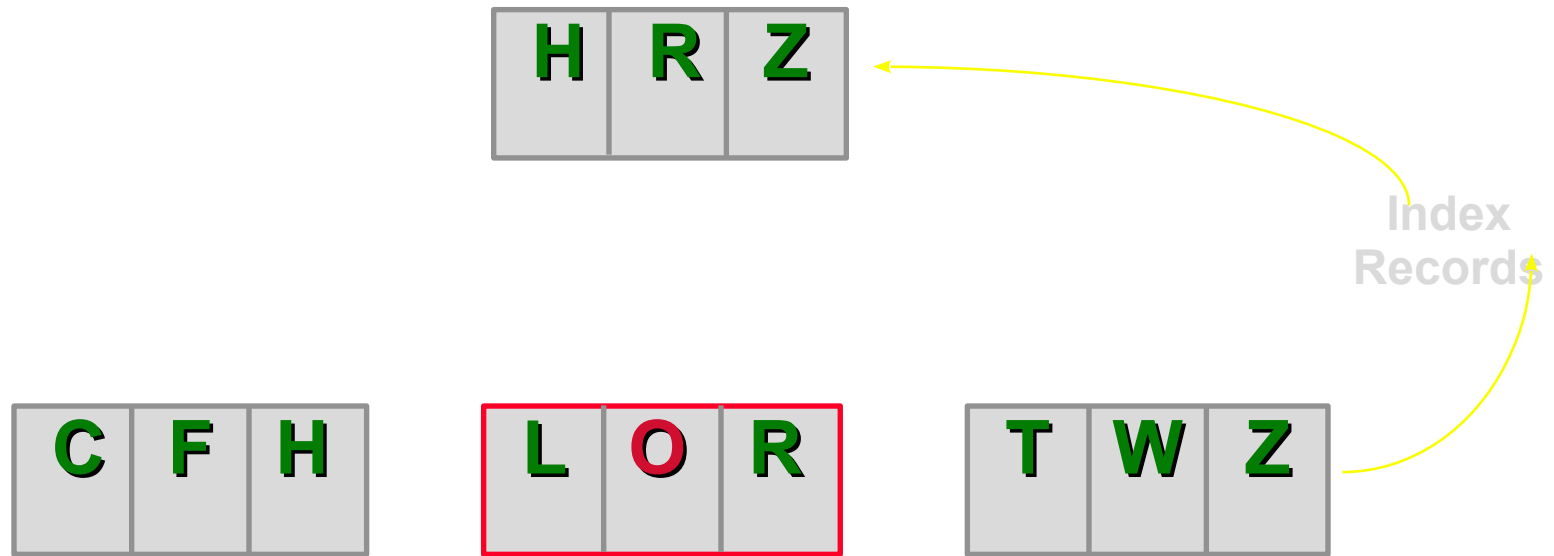
Indexed Files - Organization



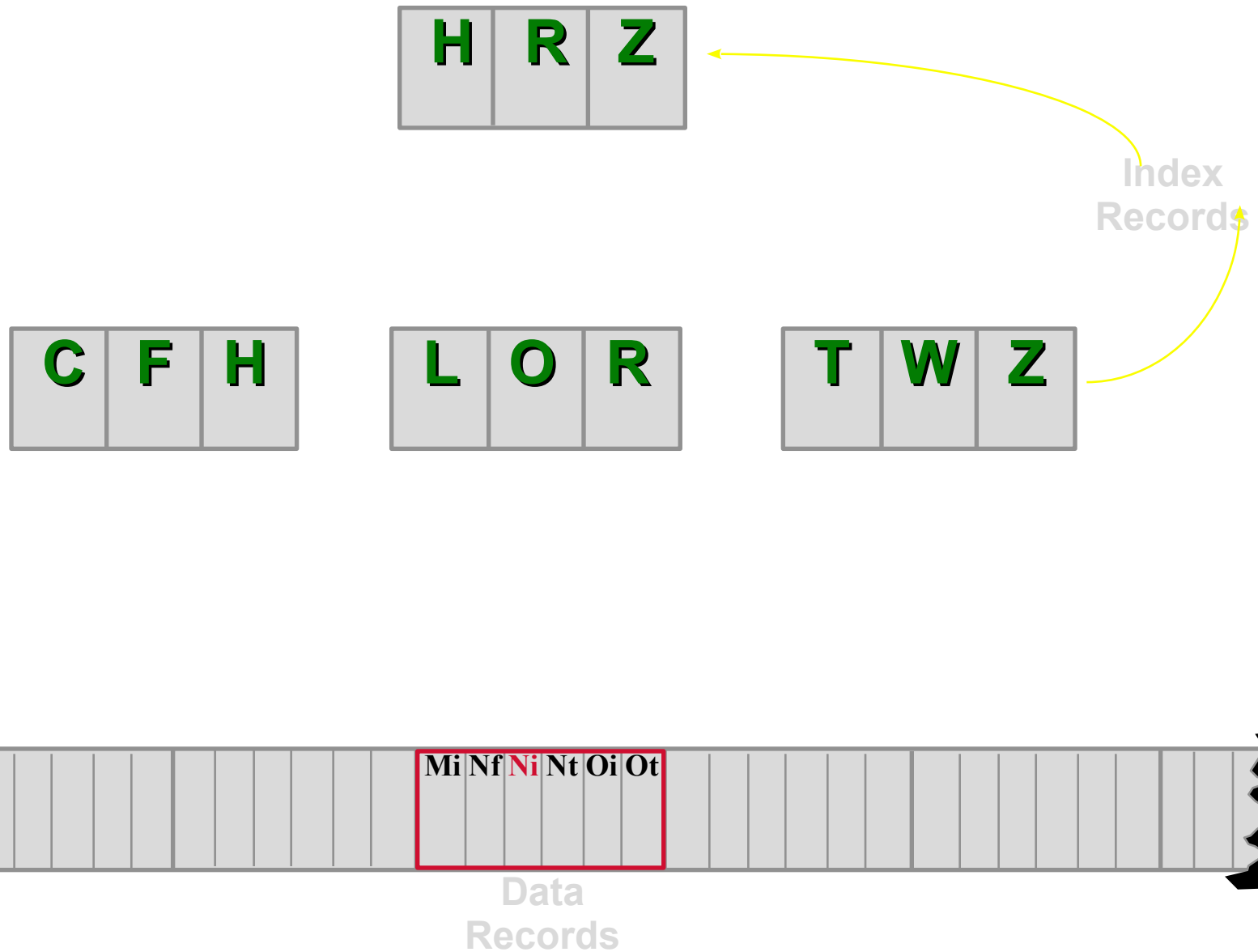
Indexed Files - Reading Record Ni



Indexed Files - Reading Record Ni



Indexed Files - Reading Record Ni



Sequential Files.

Disadvantages.

Slow - when the hit rate is low.

Complicated to change (insert, delete, amend)

Advantages.

Fast - when the hit rate is high.

Most **storage efficient**.

Simple organization.

Recovers space from deleted records.



Relative Files.

Disadvantages.

Wasteful of storage if the file is only partially populated.

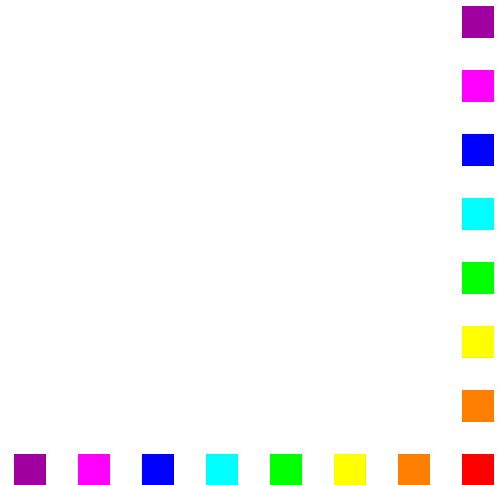
Cannot recover space from deleted records.

Only a **single, numeric key** allowed.

Keys must map on to the range of the Relative Record numbers.

Fastest Direct Access organization.
Advantages.
Very little storage overhead.

Can be read sequentially.



Indexed Files.

Disadvantages.

Slowest Direct Access organization.

Especially slow when adding or deleting records.

Not very storage efficient. Must store the Index records, the alternate Index records, the data records and the alternate data records.

Advantages.

Can use **multiple, alphanumeric keys.**

Can have **duplicate alternate keys.**

Can be **read sequentially** on any of its keys.

Can **partially recover space** from deleted records.

